

Problem Statement

Evaluate the following limit:

$$\lim_{n \rightarrow \infty} \frac{(\sum_{r=1}^n r^2) (\sum_{r=1}^n r^7)}{(\sum_{r=1}^n r^4) (\sum_{r=1}^n r^5)}$$

Detailed Solution

This limit can be elegantly evaluated by converting the summations into Riemann sums and expressing the limit in terms of definite integrals. Let's break this down step-by-step.

Step 1: Express the sums using Riemann sum notation

Recall that the right Riemann sum for a function $f(x)$ over the interval $[0, 1]$ is given by:

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n f\left(\frac{r}{n}\right) = \int_0^1 f(x) dx$$

To rewrite the terms in our limit in this form, we can factor out the appropriate power of n from each summation. For any power $p \geq 1$:

$$\sum_{r=1}^n r^p = \sum_{r=1}^n n^p \left(\frac{r}{n}\right)^p = n^{p+1} \left[\frac{1}{n} \sum_{r=1}^n \left(\frac{r}{n}\right)^p \right]$$

Applying this transformation to each of the four summations in our expression:

$$\begin{aligned} \sum_{r=1}^n r^2 &= n^3 \left[\frac{1}{n} \sum_{r=1}^n \left(\frac{r}{n}\right)^2 \right] \\ \sum_{r=1}^n r^7 &= n^8 \left[\frac{1}{n} \sum_{r=1}^n \left(\frac{r}{n}\right)^7 \right] \\ \sum_{r=1}^n r^4 &= n^5 \left[\frac{1}{n} \sum_{r=1}^n \left(\frac{r}{n}\right)^4 \right] \\ \sum_{r=1}^n r^5 &= n^6 \left[\frac{1}{n} \sum_{r=1}^n \left(\frac{r}{n}\right)^5 \right] \end{aligned}$$

Step 2: Substitute and simplify the powers of n

Now, we substitute these expressions back into the original limit:

$$L = \lim_{n \rightarrow \infty} \frac{\left(n^3 \left[\frac{1}{n} \sum_{r=1}^n \left(\frac{r}{n} \right)^2 \right] \right) \left(n^8 \left[\frac{1}{n} \sum_{r=1}^n \left(\frac{r}{n} \right)^7 \right] \right)}{\left(n^5 \left[\frac{1}{n} \sum_{r=1}^n \left(\frac{r}{n} \right)^4 \right] \right) \left(n^6 \left[\frac{1}{n} \sum_{r=1}^n \left(\frac{r}{n} \right)^5 \right] \right)}$$

Grouping the powers of n in the numerator and denominator:

$$L = \lim_{n \rightarrow \infty} \frac{n^{11} \left(\frac{1}{n} \sum_{r=1}^n \left(\frac{r}{n} \right)^2 \right) \left(\frac{1}{n} \sum_{r=1}^n \left(\frac{r}{n} \right)^7 \right)}{n^{11} \left(\frac{1}{n} \sum_{r=1}^n \left(\frac{r}{n} \right)^4 \right) \left(\frac{1}{n} \sum_{r=1}^n \left(\frac{r}{n} \right)^5 \right)}$$

Since $n \rightarrow \infty$, the n^{11} factors cancel out completely, leaving:

$$L = \lim_{n \rightarrow \infty} \frac{\left(\frac{1}{n} \sum_{r=1}^n \left(\frac{r}{n} \right)^2 \right) \left(\frac{1}{n} \sum_{r=1}^n \left(\frac{r}{n} \right)^7 \right)}{\left(\frac{1}{n} \sum_{r=1}^n \left(\frac{r}{n} \right)^4 \right) \left(\frac{1}{n} \sum_{r=1}^n \left(\frac{r}{n} \right)^5 \right)}$$

Step 3: Convert the limits to definite integrals

Using the properties of limits, we can distribute the limit to each Riemann sum factor individually:

$$L = \frac{\left(\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \left(\frac{r}{n} \right)^2 \right) \left(\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \left(\frac{r}{n} \right)^7 \right)}{\left(\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \left(\frac{r}{n} \right)^4 \right) \left(\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \left(\frac{r}{n} \right)^5 \right)}$$

Each of these limits corresponds directly to a definite integral of the form $\int_0^1 x^p dx$:

$$\int_0^1 x^p dx = \left[\frac{x^{p+1}}{p+1} \right]_0^1 = \frac{1}{p+1}$$

Evaluating the four integrals:

- $\int_0^1 x^2 dx = \frac{1}{3}$
- $\int_0^1 x^7 dx = \frac{1}{8}$
- $\int_0^1 x^4 dx = \frac{1}{5}$
- $\int_0^1 x^5 dx = \frac{1}{6}$

Step 4: Compute the final limit

Substituting these integral values back into our limit expression:

$$L = \frac{\frac{1}{3} \cdot \frac{1}{8}}{\frac{1}{5} \cdot \frac{1}{6}}$$

Simplifying the products in the numerator and the denominator:

$$L = \frac{\frac{1}{24}}{\frac{1}{30}}$$

Multiplying by the reciprocal of the denominator gives:

$$L = \frac{30}{24} = \frac{5}{4}$$

Final Answer:

$$\frac{5}{4}$$